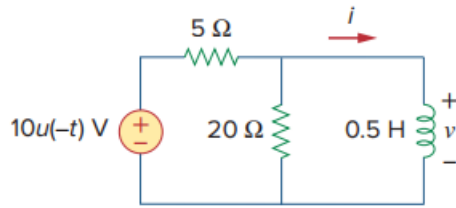


Problem

Obtain $v(t)$ and $i(t)$ in the circuit of Fig. 7.128.

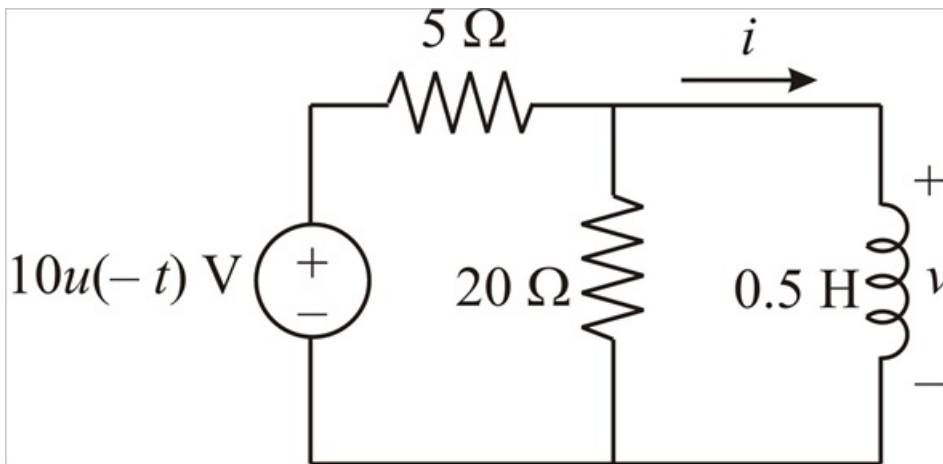
Fig. 7.128:



Step-by-step solution

Step 1 of 12

Given circuit diagram is



Step 2 of 12

We know that unit step function is defined for positive values of ' t ' and zero for negative values of ' t '

i.e.

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

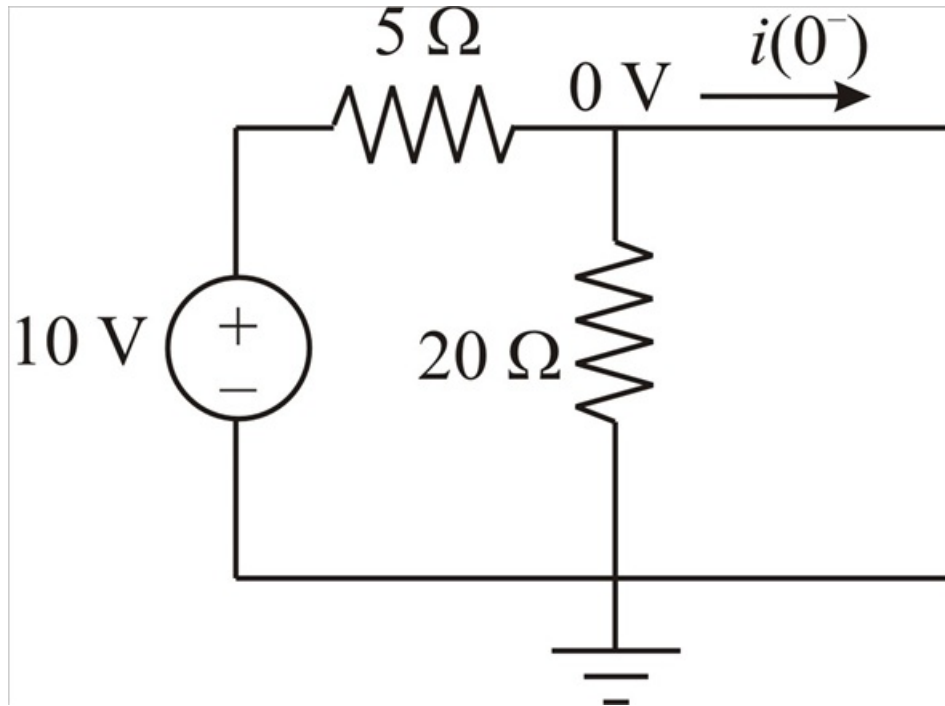
From the given circuit source is $10u(-t)$ V. This is defined for negative values of t and zero for positive values of t .

That is,

$$u(-t) = \begin{cases} 1 & t < 0 \\ 0 & t > 0 \end{cases}$$

Step 3 of 12

For $t < 0$ the voltage source is $10u(-t) = 10 \text{ V}$. At steady state, the inductor acts as a short circuit. The circuit becomes:



Step 4 of 12

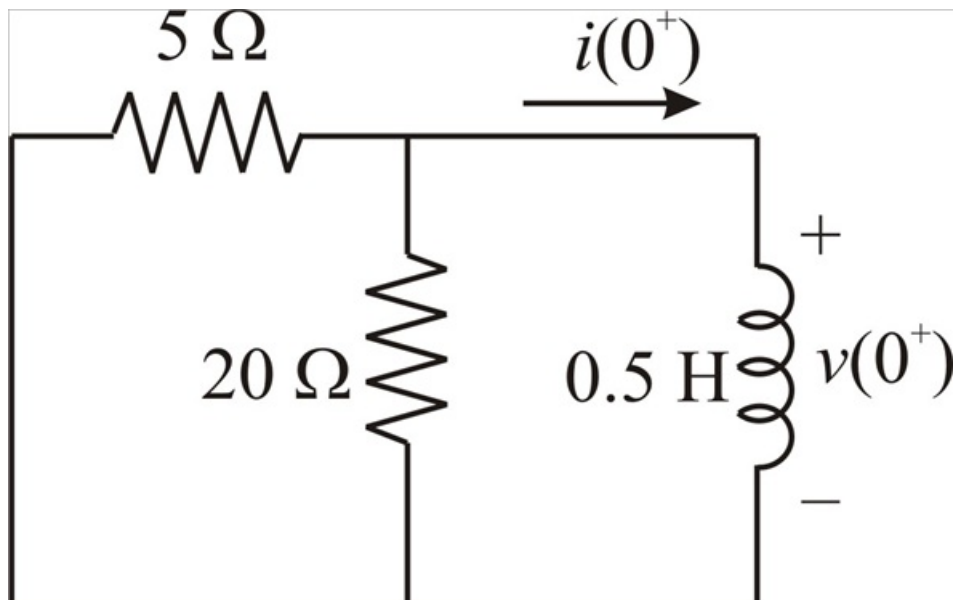
Using Kirchhoff's current law to the above circuit, the initial inductor current becomes

$$\begin{aligned} i(0^-) &= \frac{10 - 0}{5} + \frac{0}{20} \\ &= \frac{10}{5} \\ i(0^-) &= 2 \end{aligned}$$

Therefore, the initial inductor current is, $i(0^-) = 2 \text{ A}$

Step 5 of 12

For $t > 0$, the voltage source is $10u(-t) = 0 \text{ V}$. The circuit becomes



Step 6 of 12

The inductor current cannot be changed instantaneously.

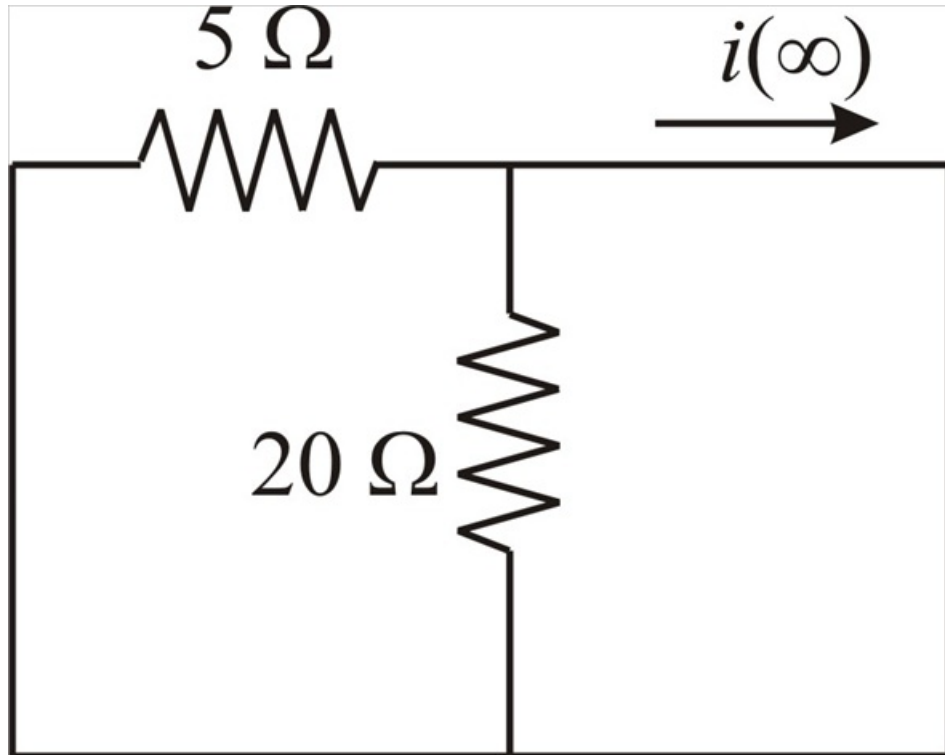
That is,

$$i(0^-) = i(0^+)$$

$$\Rightarrow i(0^-) = i(0^+) = 2 \text{ A}$$

Step 7 of 12

For $t \rightarrow \infty$. The inductor acts as a short circuit. The circuit becomes



Step 8 of 12

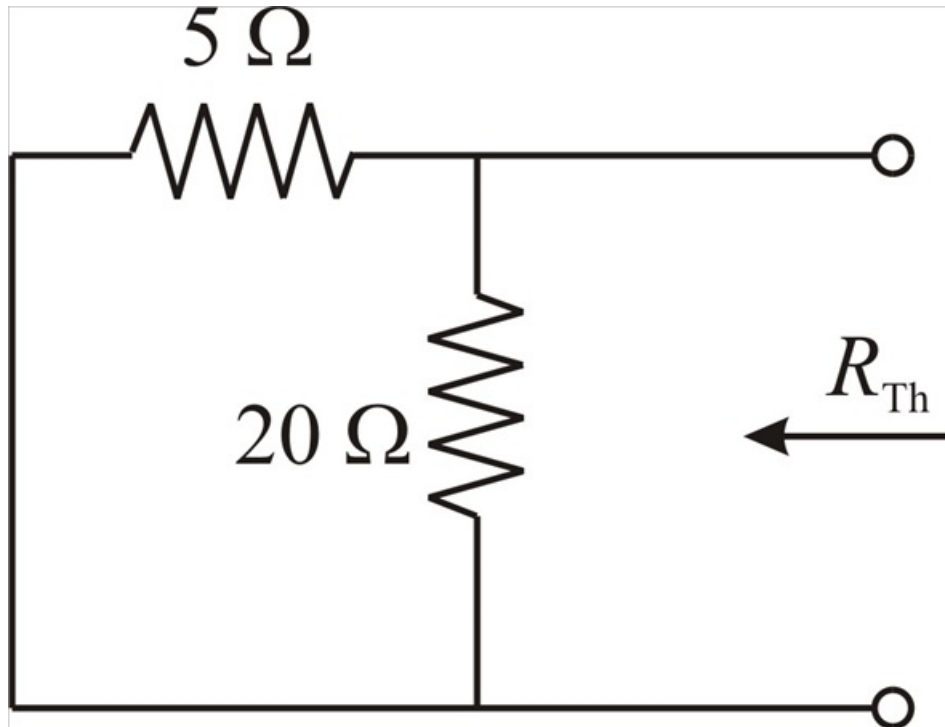
Above circuit is a source free circuit. So the final inductor current is zero.

That is,

$$i(\infty) = 0 \text{ A}.$$

The time constant is $\tau = \frac{L}{R_{Th}}$

For finding R_{Th} , consider the below circuit



Step 9 of 12

From the above circuit, the equivalent resistance is

$$\begin{aligned} R_{Th} &= 20 \parallel 5 \\ &= \frac{(20)(5)}{20 + 5} \\ &= \frac{100}{25} \\ R_{Th} &= 4 \end{aligned}$$

Therefore, the equivalent resistance is $R_{Th} = 4 \Omega$.

Step 10 of 12

Thus, the time constant is

$$\begin{aligned} \tau &= \frac{L}{R_{Th}} \\ &= \frac{0.5}{4} \\ \tau &= 0.125 \end{aligned}$$

Therefore, the time constant is $\tau = 0.125 \text{ s}$.

Step 11 of 12

The complete response of the RL circuit is

$$i(t) = i(\infty) + [i(0) - i(\infty)]e^{-\frac{t}{\tau}}$$

Substitute $i(0)$, $i(\infty)$ and τ in the above equation

$$\begin{aligned} i(t) &= 0 + [2 - 0]e^{-\frac{t}{0.125}} \\ &= 2e^{-\frac{t}{0.125}} \\ i(t) &= 2e^{-8t} \end{aligned}$$

Therefore, the inductor current is $i(t) = 2e^{-8t} u(t) \text{ A}$.

Step 12 of 12

The inductor voltage is

$$v(t) = L \frac{di(t)}{dt}$$

Substitute L and $i(t)$ in the above equation

$$\begin{aligned} v(t) &= (0.5) \frac{d(2e^{-8t})}{dt} \\ &= (0.5)(2(-8e^{-8t})) \\ v(t) &= -8e^{-8t} \end{aligned}$$

Therefore, the inductor voltage is $v(t) = -8e^{-8t} u(t) \text{ V}$.